something—a something substantial enough to provide the ultimate, absolute benchmark for motion. To many it has seemed absurd, or at least scientifically irresponsible, to base an understanding of motion on something so thoroughly imperceptible, so completely beyond our senses, that it borders on the mystical. Yet these same physicists were dogged by the question of how else to explain Newton's bucket. Mach's insights generated excitement because they raised the possibility of a new answer, one in which space is not a something, an answer that points back toward the relationist conception of space advocated by Leibniz. Space, in Mach's view, is very much as Leibniz imagined—it's the language for expressing the relationship between one object's position and another's. But, like an alphabet without letters, space does not enjoy an independent existence.

Mach vs. Newton

I learned of Mach's ideas when I was an undergraduate, and they were a godsend. Here, finally, was a theory of space and motion that put all perspectives back on an equal footing, since only relative motion and relative acceleration had meaning. Rather than the Newtonian benchmark for motion—an invisible thing called absolute space—Mach's proposed benchmark is out in the open for all to see—the matter that is distributed throughout the cosmos. I felt sure Mach's had to be the answer. I also learned that I was not alone in having this reaction; I was following a long line of physicists, including Albert Einstein, who had been swept away when they first encountered Mach's ideas.

Is Mach right? Did Newton get so caught up in the swirl of his bucket that he came to a wishy-washy conclusion regarding space? Does Newton's absolute space exist, or had the pendulum firmly swung back to the relationist perspective? During the first few decades after Mach introduced his ideas, these questions couldn't be answered. For the most part, the reason was that Mach's suggestion was not a complete theory or description, since he never specified *how* the matter content of the universe would exert the proposed influence. If his ideas were right, how do the distant stars and the house next door contribute to your feeling that you are spinning when you spin around? Without specifying a